

Auditory hardware options and budget

Computer-based sound setup and comparison with the supplied NI/TDT slides

Yes: a computer, wired powered speaker and measurement microphone can support a much cheaper setup for selected audible tones. The large NI rack is one implementation. For cortical measurements, we still use the lab’s neural recorder and connect sound timing to its recording timeline.

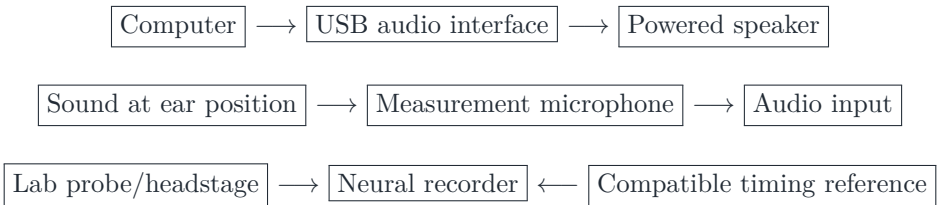
The lowest priced three-part audio basket found is **\$264.90 before accessories, delivery and tax.** About **\$425** is a conditional starting budget if the lab supplies the computer, neural recorder, a sound-level reference and compatible timing inputs. A practical audible-sound working budget is **\$1,100–\$1,600.**

The budget choices

Option	Low, USD	High, USD	Main assumption
Minimum audible	424.90	424.90	Existing PC/recorder; borrowed level reference and compatible timing connection.
Practical audible	1,114.90	1,596.87	Existing PC/recorder; allowances for calibration access and timing.
USB higher-frequency	11,004.90	20,251.98	USB NI replaces rack; retain ES1/ED1 and a suitable microphone.
Full slide rack	15,800.49	29,469.33	Both NI modules and rack links, plus complete acoustic support.

For discussion: reserve about \$1,500 for an audible first setup, or about \$30,000 for the higher-cost slide architecture. The latter is a planning scenario, not a guaranteed maximum. Unquoted items or missing recording equipment can exceed it.

How the basic system fits together



The microphone records sound. The lab probe and recorder record neural voltages. A computer playback command is not a measurement of when sound reached the ear.

What these budgets cover

All figures are USD for **one sound stimulus channel**. They exclude new probes, headstages, neural recorders, animal preparation, institutional services and integration labor. Using the lab’s own probes is the established scope; exact recorder/accessory availability remains open. The low and high columns use different purchase assumptions, not statistical bounds. **Choose one basket; do not add the alternatives together.**

The economical audible setup

One speaker, one microphone and one interface

Minimum parts list

Part	Qty	USD	Purpose and price status
Behringer UMC204HD	1	97.98	New eBay listing, Sweetwater. USB playback and analog microphone input.
JBL 305P MkII	1	99.99	Used single powered speaker. Seller says functional; no returns. Amplifier is built in.
Dayton EMM-6	1	66.93	New eBay listing, Parts Express. Analog measurement microphone.
Three-part equipment		264.90	Before accessories, shipping and tax.
Computer and neural recorder	Reuse	0.00	Incremental-cost assumption; lab availability and timing inputs to confirm.
Cables and fixed mounts	1 set	80.00	Allowance for XLR, speaker/USB wiring and repeatable geometry.
Delivery and tax reserve	1	80.00	Allowance, not a Houston shipping quote or a tax rate.
Conditional minimum		424.90	Borrow a suitable level reference and use an already compatible timing connection.

A practical budget with fewer borrowed functions

Component or allowance	Low, USD	High, USD
Interface: UMC204HD or MOTU M4	97.98	279.95
JBL speaker plus EMM-6 microphone	166.92	166.92
Cables, fixed mounts and physical loopback	100.00	150.00
Sound-level reference access/support	300.00	400.00
Recorder timing interface	250.00	350.00
Equipment and support subtotal	914.90	1,346.87
Delivery, tax and uncertainty reserve	200.00	250.00
Planning total	1,114.90	1,596.87

The high column substitutes one new MOTU M4 at \$279.95 for the UMC; it does not add a second interface. Reference access may mean borrowing or rental. Its allowance does not guarantee a new traceable calibrator or a full calibration service. A recorder needing another acquisition device can exceed the timing allowance.

Frequency limits and a smaller acoustic-only alternative

The JBL is specified at 49 Hz–20 kHz, ± 3 dB; the EMM-6 at 18 Hz–20 kHz. The UMC supports phantom-powered microphone inputs and up to 24-bit/192 kHz sampling. That sample rate does not extend the speaker/microphone's specified acoustic band.

If a wired speaker and PC already exist, a UMIK-1 USB mic at \$148.95 is a smaller acoustic-development purchase. The listing displayed \$12.95 shipping, destination dependent. Its separate clock means it is not, by itself, a synchronized cortical-recording setup. See the timing and calibration guidance on page 6.

The parts shown in the slides

One each; eBay asking prices before shipping and tax

The slides show a desktop linked to an NI rack, followed by a TDT electrostatic sound chain. Their Neuropixels link concerns separate neural acquisition. None of these listings establishes Rice inventory or a complete compatible bundle.

Item and listing	USD	Role and important dependency
NI PXIe-1082	2,650.00	Powers/holds NI modules. Used overseas example; domestic example \$6,495.
NI PXIe-6363	2,495.36	Analog/digital I/O and timing. Used; seller claims tested.
NI PXI-4461	571.33	Sound input/output. Used; hybrid-compatible version is not established.
NI PXIe-8360	493.20	Controller in chassis. Correct PXIe model required.
NI PCIe-8361	197.00	Card in desktop PC. New seller claim; verify exact identity and a free PCIe slot.
MXI 779500-03	89.85	3 m desktop-to-chassis cable. New seller claim.
SHC68-68-EPM	89.99	Used 2 m DAQ-to-breakout cable. One connector bank.
BNC-2090A	613.77	Used BNC breakout. Seller claims working; no returns displayed.
NI items subtotal	7,200.50	Conditional listing collection; compatibility and calibration remain open.
ED1/power bundle	209.06	Used ED1, HB7, PS25F and zBus chassis. Seller claims tested; fair cosmetics/rust. No ES1/accessories.
ES1 plus matched cable	1,500–2,500	Estimate. No reliable eBay speaker price found; quotation or borrowing required.

Parts that can disappear in the inexpensive path

The computer-based audible setup does not need the NI chassis, either MXI card, MXI cable, PXI sound card, NI-specific breakout/cable, ED1 or ES1. It replaces their functions with suitable audio I/O, a powered speaker and a verified recorder timing path. A standalone microphone or computer alone does not replace neural acquisition.

What is missing from the diagram

Microphone and calibration. The higher-frequency budgets add a complete Avisoft 40026 microphone kit at \$2,580 and calibration/support allowances. Its existing NI device acts as the acoustic ADC.

ED1 power. The \$209.06 surplus bundle includes compatible zBus power. Do not also purchase a second power chassis. If that bundle is unavailable, the high scenario reserves \$3,500 for replacement ED1/power; this is an estimate, not a new-unit quotation. A \$495 parts/not-working bundle is excluded.

Neuropixels. The linked brochure names PXIE_1000 and CBL_1000, with probes/headstages and chassis/interface dependencies. No new Neuropixels purchase is assumed; use the lab recording chain.

Complete higher-frequency budgets

Two architectures, each with lower and higher purchase assumptions

A new NI USB-6361, 781442-01, at \$3,030.42 can supply waveform output, acoustic acquisition and events without the rack. This candidate retains the ES1/ED1 sound chain. It is not a tested software drop-in.

Cost component	USB candidate, USD		Full slide rack, USD	
	Low	High	Low	High
NI output/acquisition chain	3,030.42	3,030.42	7,200.50	11,045.50
ED1 plus compatible power	209.06	3,500.00	209.06	3,500.00
ES1 plus matched cable	1,500.00	2,500.00	1,500.00	2,500.00
Avisoft 40026 complete kit	2,580.00	2,580.00	2,580.00	2,580.00
Band-appropriate calibration	1,500.00	3,000.00	1,500.00	3,000.00
Wiring, mounts, timing, mic power	750.00	1,500.00	750.00	1,500.00
Computer/storage: reuse or allowance	0.00	1,500.00	0.00	1,500.00
Equipment/support subtotal	9,569.48	17,610.42	13,739.56	25,625.50
Reserve at 15% of subtotal	1,435.42	2,641.56	2,060.93	3,843.83
Planning total	11,004.90	20,251.98	15,800.49	29,469.33

What changes between the columns

- The full-rack high column replaces the \$2,650 chassis with the \$6,495 domestic listing. All other NI item prices stay the same.
- The low columns use the surplus ED1/power bundle. The high columns use the \$3,500 replacement allowance. The ES1/cable, calibration and support figures are estimates in both columns.
- Both low columns reuse a suitable computer/storage setup. The high columns reserve \$1,500; the full rack needs a compatible desktop with an available PCIe slot.
- The 15% reserve covers shipping, tax, uncertainty and minor omissions. It is not a tax rate. Do not add displayed non-Houston shipping again unless replacing the reserve with actual charges.

Count the microphone kit once

CM16/CMPA48AAF-5V, part 40026, includes microphone, polarization, adjustable preamp, adjustable anti-aliasing filter, cables, clamp and mini tripod. It needs external **5 V, 52 mA**. Its stated band is **2–200 kHz**. Match input voltage/gain, anti-alias filter and sample rate to the selected NI acquisition task.

Do not add another preamp/polarization unit or the separate \$8,160 complete 116Hm recorder kit to this basket. The selected NI device is already counted as the acoustic ADC. Support allows for power and connection parts still missing.

Why this is not a fixed upper limit

No verified ES1 eBay speaker price was found. Speaker/driver quotations, calibration scope, unavailable used variants or missing recorder hardware may change the totals. An all-new rack or specialist metrology chain can cost more. These higher-frequency baskets also exclude sounds below the ES1's 4 kHz operating band.

Frequency and compatibility limits

The usable band belongs to the complete chain

The inexpensive audible basket

The specified JBL/EMM-6 bands extend to about 20 kHz. Select tones within the measured usable band and verify their level and distortion at the intended position. A 192 kHz sampling setting is not evidence that this speaker/microphone can deliver or measure ultrasound accurately.

The ES1 needs a different kind of driver

TDT ES1 is a free-field electrostatic speaker specified at **4–110 kHz, ± 11 dB**. The related EC1 is a coupler/tubing option, not the same free-field arrangement. TDT cautions that signals below 4 kHz cause substantial harmonic distortion and can degrade these speakers. Neither is a default low-frequency sheep speaker.

A computer or ordinary USB interface cannot directly drive ES1. The pictured chain needs ED1 and compatible zBus power, plus the matching speaker cable. Some integrated TDT systems, including RZ6, provide electrostatic drive themselves. Do not substitute a conventional amplifier or arbitrary speaker cable. Cable length affects response.

The slide's audio card limits the top frequency

PXI-4461 provides two analog output channels at up to **204.8 kSa/s**. At that rate, NI specifies flat output through **92.1 kHz**; the Nyquist boundary is 102.4 kHz. The pictured source therefore does not cover the ES1's entire 110 kHz specification. USB-6361 has a higher-rate waveform source, but its complete analog chain still needs measurement.

Used PXI hardware requires exact-model checks

- **PXI-4461 version:** NI sold standard PXI and hybrid-compatible versions under the same name. Only the appropriate hybrid variant fits the PXIe-1082's hybrid peripheral slots. The listed cheap card's rear connector/version is unconfirmed. [NI variant explanation](#); [chassis manual](#).
- **Controller:** use PXIe-8360 with PCIe-8361 and the compatible MXI cable. PXI-8360 is a different controller-slot architecture. [NI connection guide](#).
- **Breakout:** one SHC68-68-EPM/BNC-2090A set exposes one bank. A second is needed only if the final channel map uses both. Confirm pinouts and available channels. [NI cable/accessory guide](#).
- **Computer/software:** check OS/driver support and a free desktop PCIe slot. An existing laptop does not directly accept PCIe-8361. NI-DAQmx supports text-language programming; a LabVIEW purchase is not inherently required. [NI software options](#).

The full SONIC band needs another speaker choice

SONIC's initial alphabet spans approximately **1.4–32 kHz**; its refined set extends through 27 kHz. The 20 kHz audible basket omits the high end; ES1 omits the low end. SONIC used **USB-6361, SA1 and MF1**, not the slide's ES1 chain. For the full paper range, retain the MF2/SA1 candidate or another measured compatible route, with a microphone suitable below 2 kHz. See SONIC bioRxiv v1, Methods 6.1–6.2, p.11.

Calibration, timing and our next decisions

What turns playback into a useful neural experiment

Measure actual sound, not only the command

A software playback timestamp and an audio interface's internal digital loopback do not measure DAC output, speaker response or sound arrival at the ear. The computer, acoustic interface and neural recorder may also have independent clocks. Even modules in one rack need an explicit synchronization plan.

An economical candidate records the analog EMM-6 and an electrical playback monitor through the same audio interface. Reserve an independent physical output for a waveform/reference accepted by the neural recorder's auxiliary analog input, or use a properly conditioned marker for a compatible digital input.

Audio line output is not TTL. The recorder model determines allowable voltage, impedance, polarity, connector and grounding.

Record repeated shared markers or an appropriate analog reference, then measure acoustic onset and clock drift. A USB microphone such as UMIK-1 is convenient for calibration but does not automatically share the playback or neural clock. Psychtoolbox playback timing; physical latency test; Open Ephys synchronization; REW clock mismatch.

Establish sound levels for the full gain chain

The EMM-6's individual frequency-response file does not alone establish absolute sound pressure level after microphone-preamp gain changes. Use a suitable SPL reference and keep gains fixed. Measure each chosen tone at the intended ear position. A one-frequency reference does not establish broadband ultrasonic accuracy. REW calibration guidance.

Disable microphone input monitoring to avoid feedback. Disable automatic gain, noise suppression and sound enhancement. Save the measured response, gain, distance/orientation, background sound, sequence, configuration and timing checks with the recordings.

What we need to decide

1. **Choose the frequency range.** Selected audible tones can justify the \$1,500 first-setup budget. Full SONIC or rodent ultrasound requires a wider-band speaker and measurement chain.
2. **Identify the lab recorder.** Confirm its exact probe/headstage interface, spare analog/digital inputs, clock, supported signal levels and exported timestamps.
3. **Inventory reusable equipment.** Check the computer, wired speakers, microphones, calibrated references, DAQs, TDT power/driver/cables and fixed mounts before buying duplicates.
4. **Verify the selected chain on the bench.** Check per-tone SPL and distortion, repeatable geometry, physical acoustic onset, timing offset and drift. A price list does not establish these results.

Rice scope and the paper remain separate

Rice's primary measurements are planned in **sedated sheep**; SONIC recorded **awake, alert sheep**. Hardware feasibility does not establish response windows, decoding accuracy or achieved information-transfer rate under sedation. Lan does not require a soundproof box, behavioral rig or sound isolation; background sound, calibration and geometry still matter. The browser walkthrough is an educational simulation, not hardware control or an animal-ready protocol.

Sources and price interpretation

Direct links are clickable throughout the tables and text

How to read the prices

- **eBay asking price:** the seller’s displayed amount when researched on October 9, 2026. Not a completed-sale price, Rice quotation, reservation or independent functional/calibration test.
- **Manufacturer/public price:** Avisoft 40026 is \$2,580 manufacturer USD list. NI USB-6361 781442-01 is \$3,030.42 from Newark. Package/accessory details still need checking before buying.
- **Estimate/allowance:** ES1/cable, replacement ED1/power, calibration, timing, mounts, computers and reserves are our planning inputs, not verified vendor prices.
- **Shipping and stock:** retrieved eBay pages sometimes use cached delivery dates or another country/ZIP. Houston checkout was not performed. Expired, parts-only or conflicting-model leads are excluded from the scenario totals.

Source register

Subject	Source and what it supports
Supplied slides	System diagram and equipment links. Copied unchanged from Howard’s attachments. Reference material, not confirmed Rice inventory.
Audible components	UMC204HD, JBL 305P MkII, EMM-6, UMIK-1. Manufacturer capabilities and bands. eBay links are on page 2.
Electrostatic chain	ES1/EC1, ED1, zBus power, RZ6. Frequency, cable, driver and power dependencies.
NI hardware	4461 specifications, 6363 specifications, USB-6361, 1082 manual. Rates, bands and interfaces. eBay links are on page 3.
Microphone	Avisoft 40026 specifications and USD prices. Filtered external-acquisition kit, contents, band and power.
Neural acquisition	Neuropixels systems and Open Ephys synchronization. Separate neural hardware and shared-marker alignment.
SONIC and Rice	Perkins et al., SONIC, bioRxiv v1, Methods 6.1–6.2 (p.11), 6.6 (p.13). Team clarifications: Rice sedation and scope; message dates remain unconfirmed.
Budget records	Full research note, price inputs, editable spreadsheet. Linked prices, quantities, assumptions and formulas.

Evidence boundary

No equipment connection, acoustic bench validation, neural recording, animal procedure, vendor contact or purchase is reported here. The saved SONIC v1 PDF remains unchanged, with its provenance/checksum, attribution and **CC BY-NC-ND 4.0** notice in the repository. This is a complete equipment-planning comparison, not measured Rice performance.